Centre (MTRDC), a part of the DRDO. Operating in the S-band, the system delivers 20-nanosecond pulses of frequencies of 50Hz or 500Hz with 450mW peak power. These intense bursts of microwaves 'fry' the circuitry across a wide area, disabling drone swarms. The system is compact and scalable for use on land vehicles, naval vessels, and aerial platforms. Efforts are underway to extend the kill range up to 5km.

Surya Project. Surya project is under development as a next-generation 300-kilowatt HPL system to counter ballistic and hypersonic threats within a 20km range. The system uses a hybrid electric-driven gas laser architecture incorporating advanced beam control. The project, an integral part of the Mission Sudarshan Chakra, functions as the upper-tier directed energy shield.



A Surya laser weapon that can neutralise aerial threats with an HPL beam at a range of 20km (Source: https://www.facebook.com/defencepage)

Role of the private sector

The private sector has been contributing immensely to the development and production of DEWs. Adani Defence & Aerospace (ADA) has developed a vehicle-mounted counter-drone system featuring a 30kW HEL, in cooperation with DRDO.

The system provides 'hard kill' capability within the range of 5km and threat detection up to 10km. It also provides 'soft kill' capability across a wide area. Big Bang Boom Solutions (BBBS), a startup, has developed 'Vajra Strike,' an AI-powered DEW system to counter airborne threats. BBBS is also developing the anti drone defence system for the IAF. Bharat Electronics Limited (BEL), a PSU, is producing a 'multi-kilowatt beam directing optical channel' for HPL DEWs. BEL is also developing laser dazzlers for the Indian Navy as a potent deterrent for suspicious vehicles, aircraft, and UAVs entering secured zones. BEL has further developed a 2kW system to counter drones and UAVs. Paras Defence & Space Technologies has established a critical laser source module for use in advanced beam control systems.



DRDO developed a ground-based high-power microwave system (Source: https://defencenewsindia.in)

Box 3: Active denial system (ADS)

ADS is a non-invasive system for crowd management and protection of high-security perimeters such as nuclear power plants and airfields. The system uses millimetre waves and produces a temporary heat sensation to deter crowds without causing any lasting injury.

WaveStrike system. Bengaluru headquartered Tonbo Imaging has developed WaveStrike, a next-generation HPM system to neutralise targets with pinpoint accuracy. The WaveStrike uses multi-beam Klystron to generate massive energy with a much smaller footprint. The system consumes low power, thus increasing the mobility. Using AESA radar and high-definition thermal cameras, WaveStrike detects and 'fries' threats within 3km. A higher range of 5km version is being developed. The system provides electronic warfare defence against 'hardened' threats, such as AI-driven GPS-independent drones that bypass jammers.

DEWs: Global landscape

Across the world, countries are advancing DEWs as strategic means to neutralise complex threats such as hypersonic missiles and advanced drones and aircrafts. The emergence of hypersonic systems has exposed the limits of conventional munitions. DEWs are seen as a cost-effective and operationally viable solution, offering precision, accuracy, scalability, and ability to neutralise multiple targets in quick succession. Several nations have taken initiatives to develop DEWs. Some of the significant initiatives include:

The United States. The US has emerged as the leader with its two